Research — OpenGamepadUI

Nov 4, 2026 · @Tim

Summary

OpenGamepadUI (OGUI) gives our fullscreen mode its navigation model, screen layout and library data model, but its session-level tricks are out of reach for a Flatpak app. It is ShadowBlip's GPL-3.0 gamepad-first launcher and in-game overlay, written in Godot 4 (GDScript) with a Rust GDExtension, and designed to run inside gamescope either as the session's main client or as an overlay on top of Steam. It brings itself in front of games through gamescope's Steam X11 atoms, grabs the Guide button during play through InputPlumber, and controls TDP and clocks through PowerStation.

Worth copying: stacked navigation states, focus groups that remember position, Guide-button chords, a hero-plus-carousel home, a tabbed portrait library, and one game merged from several launch providers with four art layouts. Not portable: gamescope atom control, input interception and hardware tuning, which need a compositor-session or root-daemon role.

Project facts

OGUI is actively maintained (v0.46.1, 2026-09-03) but still describes itself as very early development, and ships only as distro packages, not as a Flatpak.

Field	Value
Repo	ShadowBlip/OpenGamepadUI; docs at readthedocs
Latest release	v0.46.1, 2026-09-03 (fixed rendering on the GNOME desktop and overlay-mode plugins); last `main` commit 2026-10-05
Recent history	v0.46.0 (2026-07-25) moved to Godot 4.7.1 and Rust 1.97.1; v0.45.1 (2026-07-15) added aarch64
Licence	README says GPL-3.0-or-later; RPM spec says GPL-3.0-only

Field	Value
Engine	Godot 4.7 (Forward+), GDScript; Rust GDExtension (godot-rust 0.4.5, tokio, zbus) for gamescope, D-Bus clients, PTY, subreaper, VDF
Required runtime	gamescope, InputPlumber, X11 libs, vulkan-tools, mesa-utils; optional PowerStation, firejail, bubblewrap, NetworkManager, BlueZ
Platforms	ChimeraOS, SteamOS (via systemd-sysext), Bazzite, Manjaro, NixOS
Packaging	AUR (`opengamepadui-bin` , `-git` , session package), SteamOS installer, NixOS module, tarball, RPM spec; no Flatpak
Session package	`gamescope-session-opengamepadui` ; ShadowBlip's repo archived 2026-02-09, Makefile now pulls OpenGamingCollective forks
Hardware profiles	Steam Deck, ROG Ally, AYANEO, AYN, AOKZOE, Anbernic, GPD, OneXPlayer, plus an AMD/Intel APU database

Runtime and session model

OGUI runs inside gamescope with two XWayland servers, poses as Steam to gamescope, and swaps focus between itself and the game by writing gamescope's control atoms.

Start-up

- `/usr/bin/opengamepadui` checks for a running gamescope; if none, it wraps itself in a nested `gamescope -e -w 1920 -h 1080 -f --xwayland-count 2 -- …` on the desktop.
- `ogui://` URIs go to a named pipe under `/run/user/$UID/opengamepadui/` ; the desktop file registers `x-scheme-handler/ogui` .
- On ChimeraOS, `chimera-session opengamepadui|steam|desktop` switches sessions, and OGUI's Power menu gains Switch to Desktop and Switch to Steam.
- The entrypoint acts as a supervisor that applies signed update packs and re-runs itself.

Two XWaylands, posing as Steam

- Gamescope runs with `--xwayland-count 2`; OGUI sorts displays into PRIMARY (carries focus atoms), OGUI (its own) and GAME; games start with `DISPLAY` set to the game XWayland.
- OGUI sets `STEAM_GAME=769` on its window (gamescope treats 769 as Steam), sets `STEAM_INPUT_FOCUS=1` and initialises `GAMESCOPECTRL_BASELAYER_APPID` to `[413091, 769]`.

Launching and tracking

1. `LaunchManager.launch()` applies per-game command, args, cwd and env overrides, sets `OGUI_ID=<name>` in the game's environment, and can wrap in firejail or bubblewrap.
2. The game runs under a Rust reaper with `PR_SET_CHILD_SUBREAPER`, so child processes stay attached.
3. OGUI pushes the `in_game` state, records the game in a recents list (`launcher.json`, up to 30), and runs the library's pre-launch, launch and exit hooks.
4. Focus: `RunningApp.grab_focus()` writes the app ID into `GAMESCOPECTRL_BASELAYER_APPID` and the window into `GAMESCOPECTRL_BASELAYER_WINDOW`; new game windows take focus automatically.
5. Tracking: on every gamescope focus change, OGUI lists windows on the game XWayland (ignoring those of 20 px or less), maps them to PIDs and groups them by `OGUI_ID`. Games started elsewhere are detected from the focused app ID.
6. App states: STARTED → RUNNING → MISSING_WINDOW → STOPPING → STOPPED. Pause sends SIGSTOP/SIGCONT; Exit sends TERM, then KILL on a second press.

Overlay over a running game

- Entering `in_game` hides OGUI's background and menus so its transparent window shows the game; InputPlumber intercept goes to PASS and `STEAM_INPUT_FOCUS=0`.
- Opening the main menu or QAM leaves `in_game`: intercept goes to ALL and input focus returns to OGUI, and menus draw over the live game. Gamescope blur is optional and off by default.

Overlay mode (`--overlay-mode`)

- Starts an underlay program (Steam in gamepad UI mode by default) and offers only QAM, settings and gamepad screens; only plugins tagged `quick-bar` load.

- InputPlumber stays in PASS with a Guide+East trigger that opens the QAM (reading of the code; unconfirmed). A systemd user unit exists for this mode; ChimeraOS calls the combination "steam-plus".

Screen inventory

OGUI's fullscreen interface is a hero-led home, a tabbed portrait library, a details screen, and two slide-in drawers: the main menu from the left and the Quick Access Menu (QAM) from the right.

Screen	Purpose	Key elements	Controller actions
Home	Resume recent games	Search ("What should you play?"), notification, help and settings icons; large hero banner of the focused card; carousel of recent games as 2:3 cards (minimum 200×300) ending in a Library tile with three random covers	D-pad moves; A opens details; B opens main menu; North focuses search
Library	Browse all games	Tabs Installed and All Games (plugins add tabs); portrait card grid remembering selection per tab; substring search	LB/RB switch tabs; A opens a game; B back
Game details	Launch, install, manage one game	Hero banner, centred logo, Play or Install button, side buttons for per-game Gamepad profile, Settings, Delete; install progress and location dialogs	A launches; B back

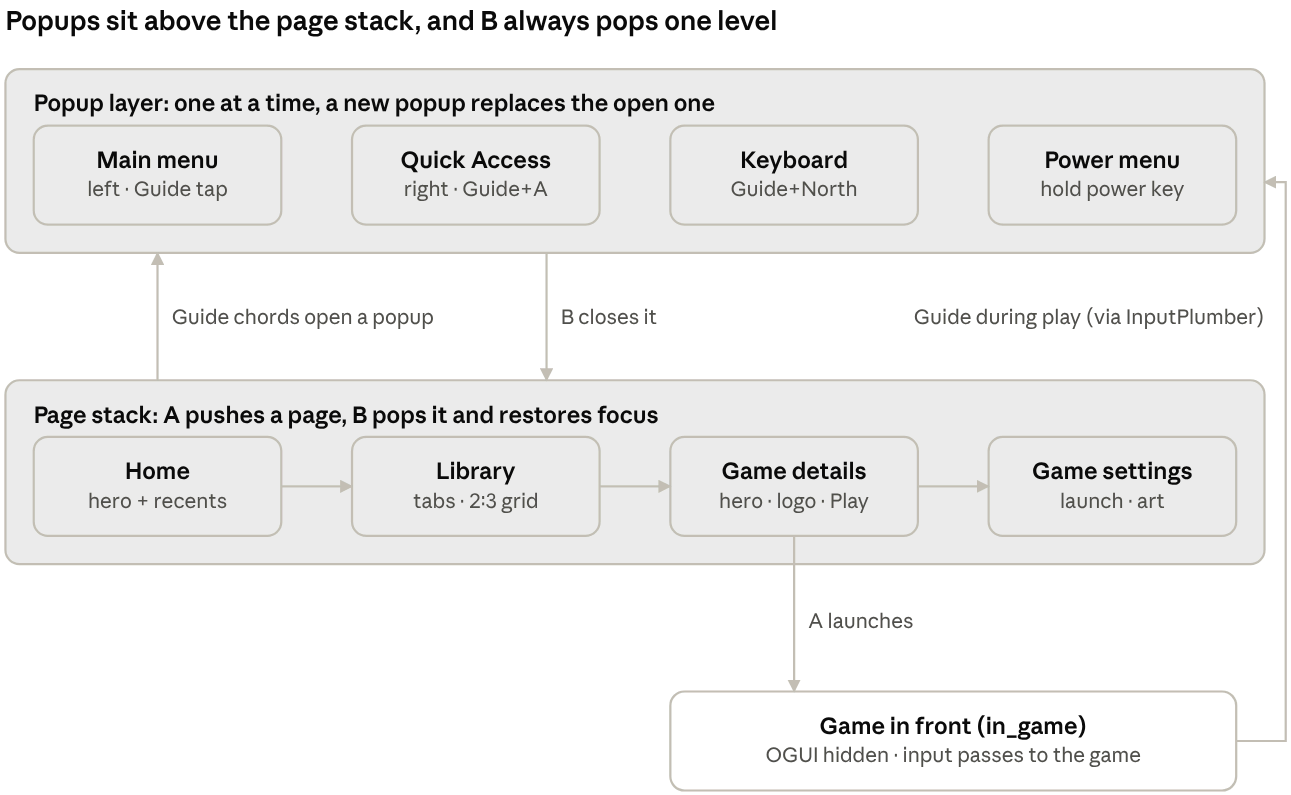
Screen	Purpose	Key elements	Controller actions
Game settings	Per-game overrides	Launch (command, args, cwd, env, provider, sandboxing), Library, Boxart (banner, grid, logo, provider)	Standard navigation
Main menu (left drawer)	Global navigation	"Playing Now" card with Return, Pause, Gamepad, Exit; then Home, Library, Store, Settings, Power	Guide tap toggles; B closes
Quick Access Menu (right drawer)	In-game quick settings	Top row: Notifications, Keyboard, Help, Gamepad, Settings; Playing Now; cards for notifications, quick settings (volume, brightness, saturation), performance, plugin cards	Guide+A or dedicated QAM button; B closes
Settings	Configuration	General, Library (desktop library toggle, recent count, hidden items), Display, Network, Bluetooth, Audio, Disks, Controllers, Plugins, Processes, Logging	LB/RB and D-pad; B back
Store	Plugin stores (for example Flathub)	Supplied by plugins	–
Power menu	Suspend, exit, switch session	Platform-specific buttons	Short power press suspends; hold opens menu
On-screen keyboard	Text entry	Layouts, shift modes; types into OGUI fields or into the game via a virtual keyboard	Guide+North or QAM Keyboard

Screen	Purpose	Key elements	Controller actions
Toasts and context bar	Feedback and hints	One toast at a time from a queue with history; bottom bar with A/B glyphs, clock, battery, "Quick Access Menu: Guide + A"	–
First boot	Setup	Language, network, timezone, plugin selection and setup	A/B

Themes are Godot theme resources (Dark Soul default, Dracula, Mountain, Water Vapor); user themes live in `~/.local/share/opengamepadui/themes`. Older screenshots show Favorites and Collections library tabs that are not in the current code.

Input and focus navigation model

OGUI drives navigation with three global state machines and focus groups that remember position; B always pops one level, and the Guide button opens popups even during play.



OGUI navigation layers · page stack, popup layer, game in front

Pages push onto a stack and pop with B; drawers, keyboard and power menu sit in a popup layer above and replace one another; during play only a Guide press, caught by InputPlumber, brings OGUI back.

State machines

- `global_state_machine`: menu, popup, in_game, in_game_menu.
- `menu_state_machine`: home, library, launcher, settings and other pages.
- `popup_state_machine`: main_menu, quick_bar, osk, power.
- Screens subscribe to entered, exited and refreshed signals to show, hide and re-grab focus; opening a popup over another uses `replace_state`.

Focus

- `FocusGroup` on a container wires focus neighbours among its children, wraps at the edges, links neighbouring groups and remembers the current focus.
- `FocusStack` keeps a stack of groups; popping restores the parent group.
- If nothing has focus when a direction is pressed, `InputManager` finds the first focusable item; Home and Library restore the last selected card.

Button mapping

Input	Action
South (A)	Accept
East (B)	Back
North	Focus search
LB / RB	Previous / next tab
L2 / R2	Trigger actions
Guide tap, released alone	Main menu
Guide held + South	Quick Access Menu
Guide held + North	On-screen keyboard
Handheld QAM button	Quick Access Menu
Volume keys	Handled globally

While Guide is held all other input is swallowed, and pressing Guide switches the controller back to the global profile so chords work inside games with remapped profiles.

Capture through InputPlumber

- OGUI sets `manage_all_devices = true`; the daemon merges physical controllers into composite devices with virtual gamepad, keyboard, mouse and touchpad targets.
- Intercept modes are NONE, PASS and ALL. In menus, ALL routes input to OGUI over D-Bus as `ui_*` events; in a game, PASS feeds the virtual gamepad, with an activation on Guide that lets the press reach OGUI (semantics inferred from code).
- Per-game controller profiles (JSON) load whenever the focused app changes.
- Without InputPlumber, OGUI falls back to Godot's own joypad and keyboard input: Ctrl+F1 main menu, Ctrl+F2 QAM, Esc back, Ctrl+PgUp/PgDn tabs. Button glyphs follow the controller type.

Library, artwork, plugins and hardware

OGUI merges launch records from several providers into one game by name, resolves four art layouts through prioritised providers, and extends everything through plugins.

Library data model

Type	Role	Key fields and methods
`Library`	One provider (Steam, desktop, plugin) registered with `LibraryManager`	`library_id`, `store_id`, icon, `supports_uninstall`; `get_library_launch_items()`, install, update, uninstall, move with progress signals, lifecycle hooks
`LibraryLaunchItem`	One provider's record of a game	`_id`, `_provider_id`, `provider_app_id`, name, command, args, env, cwd, tags, categories, installed, hidden, metadata
`LibraryItem`	One game shown to the user	Launch items from different providers merged by name; tags, categories, hidden
`LibraryManager`	Registry and queries	Sort and filter helpers: name, installed, library, hidden

- The built-in desktop library scans `~/.local/share/applications`, `/usr/share/applications` and the NixOS path for `.desktop` files with `Categories=Game`.
- The Steam plugin drives `steamcmd` through a PTY, caches app lists as JSON, and launches with `steam -gamepadui … steam://rungameid/<id>`.
- The plugin store lists seven plugins: Steam, Flathub, Lutris, SteamGridDB, Recapture, Sunshine, AYANEO Magic Modules. No Heroic plugin was found.

Artwork

- Four layouts: `GRID_PORTRAIT`, `GRID_LANDSCAPE`, `BANNER` (hero) and `LOGO`, matching SteamGridDB's grid, hero and logo types.
- `BoxArtManager` tries the game's chosen provider first, then others in registration order, then a per-layout placeholder; local art in `~/.local/share/opengamepadui/boxart`.
- Steam CDN mapping: `library_600x900.jpg` (portrait 2:3), `header.jpg` (landscape), `library_hero.jpg` (banner), `logo.png` (logo).

- A `Cache` helper stores JSON and images per folder and key alongside an HTTP image fetcher.

Plugins

- A plugin is a zip loaded as a Godot resource pack from `user://plugins`; `plugin.json` holds id, name, version, minimum API (currently 2.0.0), entrypoint and store tags.
- Store index: `plugins.json` in `ShadowBlip/OpenGamepadUI-plugins`, SHA-256 checked, with upgrade checks.
- Plugin API: `add_library`, `add_store`, `add_boxart`, `add_to_quick_bar`, `add_overlay`, `get_settings_menu`, `unload`. The docs warn plugins run arbitrary code with OGUI's privileges.

Notifications: text, icon, timeout (default 5 s), level (low, normal, critical), optional action; queued with history.

Hardware via PowerStation

- A Rust D-Bus daemon (GPL-3.0) exposing `org.shadowblip.CPU` (per-core online), `org.shadowblip.GPU.Card` (clock min/max, manual clock) and `org.shadowblip.GPU.Card.TDP` (TDP, boost, power profile, thermal limit).
- The QAM Performance card offers performance overlay, FPS limit, refresh rate, CPU boost, SMT, core count, TDP and boost, manual GPU clocks, temperature limit and power profile.
- Profiles are stored per device, docked state and game, and applied at launch or on power-source change.

Lessons for a GTK4/libadwaita fullscreen mode

Copy OGUI's navigation, focus and layout ideas; replace its input and session machinery with what a windowed Flatpak app can actually do; leave hardware and session control out.

Adopt

1. Navigation as stacked states: a base or page stack plus a separate popup layer, where B pops, popups replace each other, and each screen re-grabs focus on entry. This maps onto `AdwNavigationView` for pages plus an overlay layer for drawers.

2. Global chords: Guide tap opens a main-menu drawer, Guide+A a quick drawer, North focuses search, LB/RB switch tabs, and a bottom context bar shows A/B hints with glyphs matching the controller.
3. Focus discipline: remember focus per container and per tab, restore it on return, fall back to the first focusable widget when focus is lost, and draw large focus rings.
4. Screen layout: home with hero banner, recents carousel of 2:3 cards and a Library tile; library with tabs over a portrait grid and search; details with hero, logo, large Play button and secondary actions. This fits an Xbox-style design.
5. Data model: one game with launch items from several providers; four art layouts (portrait, landscape, hero, logo) with provider priority and placeholders; a persisted recents list.
6. Running-app tracking: tag launched processes with an environment variable and show a Playing Now card with Resume and Exit.
7. Toasts from a queue, one at a time, with history.

Adapt

- Gamepad input: without InputPlumber, a GTK app only receives gamepad events while its window has focus, via libmanette. The Guide button cannot be caught while a game is in front. An optional InputPlumber D-Bus integration is possible where it is installed (bus name and Flatpak permissions unconfirmed).
- Launch and return: from the Flatpak, launch on the host, detect exit by process or environment tag, and come back to the details screen. There is no overlay over the game; OGUI's in-game menu becomes a "game running" state when the user switches back.
- Gamescope: the simple path is OGUI's wrapper idea (`gamescope -e -f -- <app>`) or adding the app to a gamescope session. True UI/game focus swapping needs the Steam X11 atoms on XWayland; feasible in principle, unverified, not for v1. Outside gamescope, use `gtk_window_fullscreen()`.
- On-screen keyboard: only for the app's own text fields.

Out of scope for a Flatpak app

- Gamescope session and atom control (overlay, blur, base layer, FPS limit, brightness), session switching.
- InputPlumber intercept and per-game controller remapping.
- PowerStation TDP, SMT and GPU clocks; MangoApp overlay; power-button suspend.
- Network, Bluetooth and disk settings; self-update packs; code-loading plugins.

Open questions

- How InputPlumber's intercept activation works exactly, and what a Flatpak needs to talk to it.
- Whether Favorites and Collections tabs will return to OGUI's library.
- Whether a Heroic or GOG plugin exists outside the official store.
- Whether the OpenGamingCollective forks are the official successors of ShadowBlip's archived session repos.

Sources

- ShadowBlip/OpenGamepadUI (README), Releases, v0.46.1
- OpenGamepadUI documentation
- Docs in repo (commit 0db5812, 2026-10-05): installation.md, usage.md, core_systems_and_architecture.md, plugins introduction, plugins getting started
- Source: card_ui.gd, card_ui_overlay_mode.gd, launch_manager.gd, running_app.gd, input_manager.gd, focus_group.gd, focus_stack.gd, library_item.gd, library_launch_item.gd, library.gd, boxart_manager.gd, boxart_provider.gd, plugin_loader.gd, plugin.gd, performance_manager.gd, gamescope.rs, opengamepadui launcher script, PKGBUILD
- Plugin store index (plugins.json), OpenGamepadUI-steam
- gamescope-x11-client
- InputPlumber and its documentation
- PowerStation
- gamescope-session-opengamepadui (archived), OpenGamingCollective fork
- ChimeraOS wiki: General Configuration